Four-dimensional composition: exact Gaussian blocking and one-loop matching

The four directional Gaussian defects compose exactly to the sixteen-alias kernel (5), preserving the constant-flux action coefficient at every intermediate step. For sharp product blocking, after n isotropic steps the flux covariance satisfies

$$\frac{C_{2^n}(\kappa,0,0,0)_{12,12}}{t} \geq \frac{2^n}{12}|e^{i\kappa}-1|^2, \qquad 0<|\kappa|\leq\pi. \tag{1}$$

Thus the proposed finite, nondegenerate Gaussian fixed point encounters an explicit ultraviolet obstruction for these sharp observables. The small-momentum coefficient stays t at every finite depth; the two limits fail to commute. The spatial averaging in the cited perfect gauge action changes this conclusion's hypotheses.

At one loop, Theorem 3 gives continuity of the regulated background determinant in its **background Hessian jets**, with explicit norms. Section 7 constructs the generated cubic/quartic vertices and a finite commonly subtracted matching change for one pristine step, with constants (24)–(31), within the stated formal zero-image scheme. This is distinct from a bound uniform along the entire generated-action trajectory.

Theorem 4 gives the average rate $-11\log 2/(8\pi^2)$ for $SU(3)$ under sublinear endpoint matching and accumulated remainder. The remaining estimate is depth-uniform control of the full continuum-subtracted contraction and its regulator limit, together with the interacting remainder. The sharp-observable obstruction (1) defeats inference from a nondegenerate Gaussian fixed point; it does not remove the formal one-step result. Uniform endpoint matching along the generated trajectory remains open.

Inputs and scope. Use the corrected Gaussian blocking calculation and the series/parallel note, Prop. 2 and Cor. 2_s; one-step matching, §§5–8, especially (12), (15) and the many-steps remark; and the finite Λ matching in the zero-spacing note (full-read: these sections). Gaussian claims below are exact finite lattice algebra and bulk momentum estimates. Interacting matching is formal perturbation theory with its further hypotheses stated. Continuum construction and a physical spectral gap remain separate obligations. Throughout $t = g^2 = \hbar g_{\rm cl}^2$ and $b_0 = 11N/(48\pi^2)$ for pure $SU(N)$.

1. The directional defect with three transverse planes

Fix a cut direction r and write Φ_ℓ for the column of the three transverse face fluxes. Let $C_\perp(k)$ be the transverse curl from three edges to these three faces, $(C_\perp\xi)_{jk} = d_j\xi_k - d_k\xi_j$, $d_j = e^{ik_j}-1$. Use the geometric coarse heat times and put

$$\begin{aligned} &T = \mathrm{diag}_{j<k,\ j,k\neq r}(t_{jk}), \quad E = \mathrm{diag}_{j\neq r}(t_{rj}), \quad \sigma = s(1-s),\\ &V_s = \sigma C_\perp E C_\perp^*, \quad R_s = 2T + V_s, \quad \bar\Phi_\ell = (1-s)\Phi_\ell + s\Phi_{\ell+1}. \end{aligned}$$

All matrices are per colour. For positive heat times $R_s \geq 2T > 0$, even though the curl has a kernel. The Gaussian bridge has covariance σE; convolution of its face curl with the mid-face weight replaces $2T$ by R_s. Subtracting the reference coarse action gives, up to a flux-independent determinant,

$$\begin{aligned} \mathcal{D}_r &= \sum_{\ell,k}\Big[\tfrac12\bar\Phi_\ell^* R_s^{-1}\bar\Phi_\ell - \tfrac14\Phi_\ell^* T^{-1}\Phi_\ell\Big]\\ &= -\tfrac12\sum_{\ell,k}\Phi_\ell^*[(2T)^{-1} - R_s^{-1}]\Phi_\ell - \tfrac{\sigma}{2}\sum_{\ell,k}(\Phi_{\ell+1}-\Phi_\ell)^* R_s^{-1}(\Phi_{\ell+1}-\Phi_\ell) \leq 0. \end{aligned} \tag{2}$$

The weighted parallelogram identity and periodicity prove the second line, without commuting T and V_s. For a single transverse plane this is precisely Prop. 2 / Cor. 2_s. At constant flux, $V_s(0) = 0$ and $\Phi_{\ell+1} = \Phi_\ell$, so (2) vanishes exactly.

Equation (2) gives each *reference* defect. The next elimination uses the whole effective quadratic form. This is implemented below by successive Schur complements, the Gaussian specialization of the one-step note's recursion (12).

2. The composed momentum kernel

Use a periodic lattice whose side lengths are divisible by the total block factor; remove gauge and flat-link modes. Work at nonzero momentum on the exact flux subspace. Bulk limits below are taken first at each fixed block depth. With orthonormal oriented face components, define

$$(d_1(k)A)_{\mu\nu} = d_\mu A_\nu - d_\nu A_\mu, \quad q(k)^2 = \sum_\mu |d_\mu|^2,$$
$$P(k) = \frac{d_1(k) d_1(k)^*}{q(k)^2}, \qquad C_1(k) = tP(k).$$

The nonzero singular values of d_1 are q, proving the projector formula. In particular $P_{\mu\nu,\mu\nu} = (|d_\mu|^2 + |d_\nu|^2)/q^2$.

Theorem 1 (four Schur complements, exact). Let $\mathcal{T}_r$ act on the current covariance by

$$\begin{aligned}(\mathcal{T}_r C)(K) &= \tfrac{1}{2}\sum_{\epsilon=0}^{1} F_r(k^\epsilon) C(k^\epsilon) F_r(k^\epsilon)^*,\\ k_r^\epsilon &= K_r/2 + \pi\epsilon, \quad k_j^\epsilon = K_j \ (j \neq r),\\ (F_r)_{\mu\nu,\mu\nu}(k) &= \begin{cases} 1 + e^{ik_r}, & r \in \{\mu,\nu\},\\ 1, & r \notin \{\mu,\nu\}.\end{cases}\end{aligned} \tag{3}$$

At every stage the precision is the inverse covariance on its exact range. Equivalently, after writing the gauge-reduced action in retained variables x and eliminated variables y, its update is

$$Q_{\text{next}} = Q_{xx} - Q_{xy} Q_{yy}^{-1} Q_{yx}. \tag{4}$$

After all four directions, in any order,

$$\boxed{\begin{aligned} C_2(K) &= \frac{t}{16} \sum_{\nu \in \{0,1\}^4} F(k_\nu) P(k_\nu) F(k_\nu)^*, \quad k_\nu = K/2 + \pi\nu,\\ F_{\mu\nu,\mu\nu}(k) &= (1 + e^{ik_\mu})(1 + e^{ik_\nu}), \qquad Q_2(K) = C_2(K)|^{-1}_{\operatorname{im} d_1(K)}. \end{aligned}} \tag{5}$$

Here the alias multi-index in the sum is separate from face indices. If Q_2 is extended by zero off the exact range, the effective action is $\frac{1}{2}\sum_K \Phi(K)^* Q_2(K) \Phi(K)$, and the coarse link Hessian is $d_1(K)^* Q_2(K) d_1(K)$. Formula (5) is an explicit lattice-momentum expression for the nested complements (4), retaining every generated quadratic interaction.

Proof. The factor $1/2$ in (3) is the squared unitary Fourier normalization for decimation in one direction. Stokes supplies its face factor. Gaussian pushforward gives BCB^*; completing the square gives (4). The finite gauge-reduced integrals satisfy Fubini. Multiplying the directional factors and the four normalizations yields (5). All intermediate conditional precisions are positive on the integrated subspaces. □

Constant-flux preservation at every intermediate step. Adjoin a harmonic background $F_{\mu\nu}$, or use twisted affine boundary data, with its Maxwell action. On a lattice with side lengths a_μ the constant flux is $\Phi_{\mu\nu} = a_\mu a_\nu F_{\mu\nu}$ and $t_{\mu\nu} = t a_\mu^2 a_\nu^2 / \prod_\rho a_\rho$. Consequently

$$\frac{1}{2}\sum_{x,\mu<\nu} \frac{|\Phi_{\mu\nu}|^2}{t_{\mu\nu}} = \frac{V_{\text{phys}}}{2t} \sum_{\mu<\nu} |F_{\mu\nu}|^2. \tag{6}$$

Constant fine flux is orthogonal in this action to every periodic exact fluctuation: the sum of a curl is zero. Blocking fixes its harmonic component. It is therefore the constrained minimizer for constant coarse data at every intermediate elimination, including those starting with a generated action. This proves (6) survives each of the four steps. For nonzero momenta, G1 gives the corresponding isotropic limit $C_2(K) = t[P_{\mathrm{cont}}(\hat{K}) + O(K^2)]$ in centred phases. Periodic exact forms alone have zero harmonic flux; (6) specifies the extension used to speak of a constant-flux coefficient.

3. Iteration: an explicit sharp-blocking obstruction

After n steps let $b = 2^n$ and define

$$\begin{aligned} s_b(z) &= \sum_{j=0}^{b-1} e^{ijz}, \quad F_b(k)_{\mu\nu,\mu\nu} = s_b(k_\mu)s_b(k_\nu), \\ k_m &= (K + 2\pi m)/b, \quad m \in \{0, \dots, b-1\}^4, \\ C_b(K) &= tb^{-4} \sum_m F_b(k_m)P(k_m)F_b(k_m)^*, \qquad Q_b = (C_b|_{\mathrm{im}\, d_1(K)})^{-1}. \end{aligned} \tag{7}$$

Composition of sums along links proves (7) directly. Each finite b has the same leading small-momentum coefficient t, by G1.

Theorem 2 (growth with explicit constants). The covariance (7) satisfies (1). More generally, for $K \in (-\pi, \pi]^4$ with $K_1 \neq 0$,

$$C_b(K)_{12,12}/t \;\geq\; \frac{b}{4\pi^2}|e^{iK_1} - 1|^2. \tag{8}$$

It therefore diverges on sets of positive momentum measure and in every L^p operator norm, $1 \leq p \leq \infty$. No finite covariance limit exists in these norms. On the axis of (1), $\|Q_b(K)\| \leq 12/[tb|e^{i\kappa} - 1|^2]$, so uniform convergence to a nondegenerate precision also fails.

Proof. At $K = (\kappa, 0, 0, 0)$ only $m_2 = 0$ survives in the 12 diagonal. Since $s_b(0) = b$ and $|s_b(k_1)|^2|d_1(k_1)|^2 = |e^{i\kappa} - 1|^2$,

$$C_b(K)_{12,12}/t = b^{-2}|e^{i\kappa} - 1|^2 \sum_{m_1,m_3,m_4} \frac{1}{q(k_m)^2} \geq b|e^{i\kappa} - 1|^2/12.$$

Here $q^2 \leq 12$, since $k_2 = 0$. Off-diagonal entries among 12, 13, 14 vanish: shared support forces the other two transverse momenta to zero, and their projector cross entry then vanishes. By transverse symmetry these three diagonal entries agree. They span the exact subspace on this axis, giving the precision bound.

For (8), keep just the b^3 aliases with $m_2 = 0$ in the positive diagonal sum. Use $P_{12,12} \geq |d_1|^2/q^2$, $q^2 \leq 16$, and

$$|s_b(K_2/b)| = \frac{|\sin(K_2/2)|}{|\sin(K_2/(2b))|} \geq \frac{2b}{\pi},$$

with its continuous value at $K_2 = 0$. Cancellation of $|d_1|^2$ gives $b^{-4}b^3|e^{iK_1} - 1|^2(4b^2/\pi^2)/16$, which is (8). For example on $\pi/2 \leq K_1 \leq \pi$, the bound is $tb/(2\pi^2)$; this set has normalized Brillouin measure $1/4$. Thus $\|C_b\|_{L^p} \geq tb(1/4)^{1/p}/(2\pi^2)$, with exponent zero for $p = \infty$. □

This calculation identifies the ultraviolet boundary fluctuations of a sharp plaquette surface. It also shows why G1's remainder constants cannot be uniform in b. Taking $K \to 0$ first keeps the Maxwell coefficient; taking $b \to \infty$ at fixed nonzero axial K sends the precision to zero. A topology which controls a finite nondegenerate propagator and its inverse cannot supply the proposed fixed point. Possible degenerate precision limits alone would be insufficient for one-loop determinant continuity.

Relation to the cited fixed points. Bell–Wilson's finite-lattice Gaussian RG (metadata and abstract checked) supplies the general framework. Bietenholz–Wiese's §3, equations (3.1)–(3.6) (passage; DOI verified) averages the continuum gauge field over adjoining hypercubes, with form factor $\Pi_\mu(p) = (\widehat{p}_\mu/p_\mu)\prod_\rho(\widehat{p}_\rho/p_\rho)$. Their delta constraint fixes this spatial average exactly. Sending the constraint's Gaussian width to zero retains the transverse averaging. Our link product samples a sharp line, and (7) has no such transverse factors. The distinction explains why that delta limit supplies no convergence theorem for (7).

4. A one-loop continuity theorem with the necessary norm

Let $\mathcal{K}_n$ denote the full dimensionless action shape, including vertices and any one-loop local terms, and ζ_n its anisotropy. At isotropic endpoints $\zeta_n = 1$. Fix the common background scheme of the one-step note. In that scheme, the weakest scalar boundedness requirement is $\sup_n |c(\zeta_n, \mathcal{K}_n)| < \infty$; compactness of all action parameters is stronger than necessary. For the average rate even this can be relaxed to sublinear endpoint differences (Theorem 4).

Here is a useful sufficient condition directly on the determinant. Introduce a common infrared regulator and finite volume, remove zero modes consistently, and normalize a smooth external background $\epsilon\mathcal{B}$ so its classical action per volume is $u\epsilon^2/2$. All background differentiation below is at fixed regulator, boundary prescription and lattice. Write the positive vector and ghost Hessians as

$$H_{j,n}(\epsilon) = H_{j,n} + \epsilon V_{j,n} + \epsilon^2 W_{j,n} + O(\epsilon^3), \quad j = 1, 0.$$

These are full operators, allowing momentum transfer by the background. Define their relative jets

$$A_{j,n} = H_{j,n}^{-1/2} V_{j,n} H_{j,n}^{-1/2}, \qquad E_{j,n} = H_{j,n}^{-1/2} W_{j,n} H_{j,n}^{-1/2}. \tag{9}$$

Use trace per lattice volume τ. Set $\|X\|_{1,\tau} = \tau|X|$ and $\|X\|_{2,\tau}^2 = \tau(X^*X)$; these include colour and vector indices. Let m_n contain the measure, constraint Jacobian and explicitly supplied local contributions in this normalization.

Theorem 3 (regulated background determinant). Its inverse-coupling coefficient is

$$c_n^{\mathrm{reg}} = \tau\left(E_{1,n} - \tfrac{1}{2}A_{1,n}^2 - 2E_{0,n} + A_{0,n}^2\right) + m_n. \tag{10}$$

If uniformly in n, $\|E_{j,n}\|_{1,\tau} \le M_{E,j}$, $\|A_{j,n}\|_{2,\tau} \le M_{A,j}$ and $|m_n| \le M_m$, then

$$|c_n^{\mathrm{reg}}| \le M_{E,1} + \tfrac{1}{2}M_{A,1}^2 + 2M_{E,0} + M_{A,0}^2 + M_m. \tag{11}$$

Convergence of E_j in trace norm, A_j in this Hilbert–Schmidt norm, and m_n implies convergence of c_n^{reg}. Explicitly, for two shapes n, l,

$$\begin{aligned} |c_n^{\mathrm{reg}} - c_l^{\mathrm{reg}}| \le{}& \|E_{1,n} - E_{1,l}\|_{1,\tau} + \tfrac{1}{2}(\|A_{1,n}\|_{2,\tau} + \|A_{1,l}\|_{2,\tau})\|A_{1,n} - A_{1,l}\|_{2,\tau} \\ &+ 2\|E_{0,n} - E_{0,l}\|_{1,\tau} + (\|A_{0,n}\|_{2,\tau} + \|A_{0,l}\|_{2,\tau})\|A_{0,n} - A_{0,l}\|_{2,\tau} + |m_n - m_l|. \end{aligned} \tag{12}$$

Proof. Expand $\operatorname{Tr}\log H(\epsilon)$: its ϵ^2 coefficient is $\operatorname{Tr}(H^{-1}W - \frac{1}{2}H^{-1}VH^{-1}V)$. Use $\Gamma_1 = \frac{1}{2}\log\det H_1 - \log\det H_0$ and multiply by 2 to extract the coefficient of $u\epsilon^2/2$. Cyclicity proves (10). Trace Hölder and $|\tau(A^2 - B^2)| \le (\|A\|_{2,\tau} + \|B\|_{2,\tau})\|A - B\|_{2,\tau}$ prove (11)–(12). □

For example, a family of background quadratic operators with $H_j \ge \gamma I$, $\|V_j\| \le v_j$, $\|W_j\| \le w_j$, and d_j components per site obeys (11) with $M_{E,j} = d_j w_j/\gamma$ and $M_{A,j} = \sqrt{d_j}v_j/\gamma$. In four dimensions $d_1 = 4(N^2-1)$ and $d_0 = N^2 - 1$ (32 and 8 for $SU(3)$). Operator-norm convergence of these three jets with the same positive γ proves convergence at fixed regulator. This is an actual continuity theorem for a specified Gaussian-plus-background family. The positive bridge

mass in a single pristine elimination provides such fixed-step control. Maintaining it and the jet bounds after arbitrarily many generated eliminations is a further requirement.

Removing regulators and extracting F^2. In the massless problem the individual terms in (11) can diverge with the regulator. Let $\mathcal{J}_n(k;\rho)$ be the combined vector, ghost and measure integrand for the F^2 coefficient, with the same continuum infrared subtraction at every n. Here ρ collectively denotes the infrared and external-momentum regulators; the bulk limit is already taken. Assume the background Ward identities and the extraction of this integrand, including the external-momentum derivatives, are valid. Write

$$c_n = c_{\mathrm{ref}} + \lim_{\rho\to 0} \int_{\mathcal{B}_4} \mathcal{J}_n(k;\rho)\, \frac{d^4k}{(2\pi)^4}. \tag{13}$$

The concrete sufficient boundedness condition is

$$\sup_{n,\rho} \int |\mathcal{J}_n(k;\rho)|\, \frac{d^4k}{(2\pi)^4} \le M, \tag{14}$$

together with existence of each limit in (13). It gives $|c_n| \le |c_{\mathrm{ref}}| + M$. A common integrable envelope, pointwise regulator limits and convergence of $\mathcal{J}_n$ to a limit uniformly in ρ in L^1 imply $c_n \to c_*$ by dominated convergence. Condition (14) permits cancellations within the combined integrand and is weaker than bounding every unrenormalized jet norm. Bounded scalar integrals can also occur without (14); we claim (14) as a sufficient analytic criterion, not a necessary one.

At the background level, Schur complements give an additional exact finite-depth identity:

$$\Gamma_1[H] = \Gamma_1[H_{yy}] + \Gamma_1[H/H_{yy}], \tag{15}$$

separately for vectors and ghosts with their stated weights, and with any measure terms retained. Differentiate this determinant identity twice before removing regulators. All four updated Hessians compose exactly. Equation (15) controls bookkeeping through Gaussian background elimination; it supplies no depth-uniform estimate for (14). Sharp blocking's Theorem 2 obstructs an argument based just on convergence to a uniformly invertible zero-background kernel.

5. The generated vertices that still need control

In (9), V contains the vertex with two quantum legs and one external background leg; W contains two quantum and two background legs. Thus (10) contains the cubic bubble and quartic tadpole, as well as the ghost and measure terms. For a spatially varying background the products in (10) are operator products. Their momentum expansion includes the orbital terms that a pointwise matrix determinant would miss.

At one loop higher tree vertices cannot enter the term with two external legs: the loop and leg counts force either two cubic vertices or one quartic vertex. Generated one-loop two-leg terms must be carried explicitly in m_n. A sufficient missing estimate is control of (9) in the norms of Theorem 3 after the common subtraction, or directly (14), **uniform in depth and volume and through the external-momentum/infrared limits**. This includes bounds for the external derivatives selecting F^2, the ghost vertices and the background dependence of the blocking constraint. Ward identities fix the allowed leading structure; the finite integrals need this additional quantitative estimate.

The freedom left by the quadratic kernel is concrete. In local near-identity plaquette coordinates $X_p = \log U_p$, the gauge-invariant term $\eta \sum_p |X_p|^4$ has zero quadratic and cubic part at the trivial field. Around $X = \epsilon B + f$, its quadratic fluctuation part at order ϵ^2 is

$$\eta\epsilon^2\big(2|B|^2|f|^2 + 4\langle B, f\rangle^2\big). \tag{16}$$

For $\eta > 0$ this gives a positive nonzero Gaussian tadpole. One can vary η while preserving the zero-field quadratic kernel exactly. Smooth gauge-invariant continuations away from the identity have the same Taylor coefficients. This demonstrates why kernel convergence, even if supplied by a different blocking, needs a controlled background completion before it implies continuity of c. Equation (16) describes allowed action shapes; it makes no claim that this particular η_n is generated along our trajectory.

Theorem 2 likewise establishes neither boundedness nor growth of c_n: vector/ghost/vertex cancellations could still control (14) despite a divergent sharp-flux covariance. Estimating that combined quantity, or modifying the observables to a spatially averaged blocking and proving its vertex estimates, are the two concrete next routes.

6. Average running and finite scale matching

Theorem 4 (conditional average, with remainder). Consider n isotropic coarsenings, retaining the full generated actions. Assume the one-step background-matching hypotheses at each step, including no extra massless modes and the common definition of the coupling. Put $c_k = c(\zeta_k, \mathcal{K}_k)$ and assume

$$u_{k+1} - u_k = -2b_0 \log 2 + c_k - c_{k+1} + r_k, \qquad |r_k| \le A t_k \tag{17}$$

with a common finite A. Then

$$\left| \frac{u_n - u_0}{n} + 2b_0 \log 2 \right| \le \frac{|c_0 - c_n|}{n} + \frac{A}{n} \sum_{k<n} t_k. \tag{18}$$

In particular, $|c_0 - c_n| = o(n)$ and $\sum_{k<n} t_k = o(n)$ imply the universal average. Bounded $|c_k| \le C$ gives $2C/n$ for the first term; boundedness is sufficient and strictly stronger than sublinear growth. Formal one-loop truncation sets $r_k = 0$.

For an explicit weak-coupling remainder bound, assume a family of trajectories with increasingly fine initial lattices and

$$t_k \le \frac{1}{U + \alpha(n-k)}, \qquad U > 0, \quad \alpha > 0, \quad 0 \le k < n, \tag{19}$$

where U, α, A are independent of n. Then

$$\sum_{k<n} t_k \le \frac{1}{\alpha} \log(1 + \alpha n/U), \qquad \left| \frac{u_n - u_0}{n} + 2b_0 \log 2 \right| \le \frac{2C}{n} + \frac{A}{\alpha n} \log(1 + \alpha n/U). \tag{20}$$

Proof. Sum (17); all intermediate c_k cancel. For (20), bound the decreasing sum $\sum_{j=1}^{n} (U + \alpha j)^{-1}$ by its integral from 0 to n. □

For $SU(3)$, $2b_0 = 11/(8\pi^2)$, so the average is $-11 \log 2/(8\pi^2)$. At leading matched one-loop order the expected value of α is $2b_0 \log 2$; (19) is an explicit extra hypothesis for controlling the remainders. An infinite coarsening from a fixed initial coupling eventually leaves weak coupling. The limit in (20) uses trajectories with increasingly weak initial coupling and a weak final endpoint, as in the many-steps remark. If the initial shapes also vary, the requirement is the endpoint difference in (18), rather than a statement about c_n alone. No running approximation is used to prove the bound that is supposed to justify it.

Finally, $u_B = u_A + d$ at the same scale gives $\Lambda_B/\Lambda_A = e^{-d/(2b_0)}$ at one loop. For $u_{\text{matched}} = u + c_n$, this reads $\Lambda_{\text{matched}}/\Lambda_{\text{bare,n}} = e^{-c_n/(2b_0)}$. Thus a bounded matching family has finite scale ratios in $[e^{-C/(2b_0)}, e^{C/(2b_0)}]$. This is the action-shape version of the finite Λ matching in the zero-spacing note. Defining the matched coupling absorbs the endpoint terms by definition; identifying the bare blocked coupling's average still requires (18).

7. The first generated vertices and common subtraction (formal)

One pristine isotropic midpoint step. The following are coefficients of the zero-image expansion in t; the bounds concern those coefficients, uniformly in spatial volume. Use dimensionless links, periods at least three, and the mid-vertex gauge of Proposition 1 (passage). Write the transported bridge endpoints as e^{a_e}, e^{b_e} and the mid-link as e^{y_e}. The dimensionless classical action for $-\log\Psi$ is the stationary value in y of

$$\begin{aligned} s(x,y) = \sum_e (|\log(e^{-a_e}e^{y_e})|^2 + |\log(e^{-y_e}e^{b_e})|^2 - \tfrac{1}{2}|\log(e^{-a_e}e^{b_e})|^2) \\ + \tfrac{1}{4}\sum_p |\log \prod_{e\in\partial p} e^{\epsilon_{pe}y_e}|^2, \qquad x=(a,b). \end{aligned} \tag{21}$$

The contribution to the action is s_*/t. Old transverse weights and the coarse electric action are added separately to obtain the full blocked action. For original coarse links, substitute the paths $e^a = P$, $e^b = Q^{-1}$ from Proposition 1 before taking coefficients; this also retains the background dependence of the constraint.

Here is an explicit finite-word prescription for every lattice kernel. For an ordered word with signed letters $z_1, \dots, z_m$, $m \le 4$, set

$$\begin{gathered} E_r = \sum_{n_1+\cdots+n_m=r} \frac{z_1^{n_1}\cdots z_m^{n_m}}{n_1!\cdots n_m!}, \quad F_1 = E_1, \quad F_2 = E_2 - \tfrac{1}{2}E_1^2, \\ F_3 = E_3 - \tfrac{1}{2}(E_1E_2 + E_2E_1) + \tfrac{1}{3}E_1^3, \\ p_2 = |F_1|^2, \quad p_3 = 2\langle F_1, F_2\rangle, \quad p_4 = |F_2|^2 + 2\langle F_1, F_3\rangle. \end{gathered} \tag{22}$$

Use the invariant metric $|X|^2 = -2\operatorname{tr}X^2$ on $\mathfrak{su}(N)$. In each term of (21) replace the squared logarithm by p_r, with its displayed weight, obtaining s_r. Define the symmetric kernels $Q = \partial^2 s_2$, $T = \partial^3 s_3$, $U = \partial^4 s_4$ at zero. Thus (22), signed face incidences and finite sums specify all colour and position indices, without an implicit interacting expectation. Put $H = Q_{yy} = 4I + C^*C/2$, $G = H^{-1}$, $B = Q_{yx}$ and $Lx = (x, -GBx)$; in endpoint coordinates $Bx = -2(a+b)$. The generated kernels are

$$\begin{aligned} (\Gamma_3)_{ijk} &= \sum_{ABC} T_{ABC}L_{Ai}L_{Bj}L_{Ck}, \\ Z_{e,ij} &= \sum_{AB} T_{eAB}L_{Ai}L_{Bj}, \\ (\Gamma_4)_{ijkl} &= \sum_{ABCD} U_{ABCD}L_{Ai}L_{Bj}L_{Ck}L_{Dl} - \sum_{ef} G_{ef}(Z_{e,ij}Z_{f,kl} + Z_{e,ik}Z_{f,jl} + Z_{e,il}Z_{f,jk}). \end{aligned} \tag{23}$$

Here repeated mid-edge indices include the colour pairing. The vertices are $\Gamma_3[x^3]/6$ and $\Gamma_4[x^4]/24$; their background Hessian jets are $V = \Gamma_3[\mathcal{B},\cdot,\cdot]$ and $W = \Gamma_4[\mathcal{B},\mathcal{B},\cdot,\cdot]/2$. Indeed $y = -GBx - GZ[x,x]/2 + O(x^3)$; substituting into $s_2+s_3+s_4$ gives the exchange contribution $-\langle Z[x,x], GZ[x,x]\rangle/8$.

Explicit size and decay. Let d be distance between edges in the graph joining edges of a common spatial face, attaching endpoint variables to their corresponding mid-edge. Since $4I \le H \le 10I$, the Neumann series about $7I$ gives $\|G_{ef}\| \le (3/7)^{d(e,f)}/4$. This is the three-dimensional version of Lemma 1 (passage). Put

$$\alpha = \tfrac{1}{2}\log(7/3), \quad q = \sqrt{3/7}, \quad g = 48\frac{1+4q+q^2}{(1-q)^4}, \quad K = 2^{40}(1+N)^4 e^{6\alpha}, \quad l = 1+gK. \tag{24}$$

Use the anchored kernel norm: fix any one argument, sum the operator norm over all other edge arguments, with weight $e^{\alpha\mathsf{T}}$, where T is the shortest connecting tree length. The same definition for matrices includes both row and column sums. Then

$$\|G\|_\alpha \le g, \quad \|B\|_\alpha, \|T\|_\alpha, \|U\|_\alpha \le K, \quad \|\Gamma_3\|_\alpha \le Kl^3 =: v_3, \quad \|\Gamma_4\|_\alpha \le (K + 3K^2 g)l^4 =: v_4. \tag{25}$$

For the first bound, a radius-n edge ball has at most $3(4n+3)^3$ members; sum $\frac{3}{4}\sum_n (4n+3)^3 q^n \le g$. For the local bounds, the absolute word-coefficient sums in (22) are at most 8^r for $r \le 3$; multiplication, the derivative factor 4!, the weights in (21), and at most 24 incident words are covered by K. The same constant covers the length-three endpoint paths after substitution. Tree lengths are subadditive when kernels are joined, proving (25) from (23). In particular each kernel is bounded by $v_r e^{-\alpha \mathsf{T}}$. These are deliberately loose constants; t^{-1} restores action units.

7.1. One-loop contraction after common subtraction (formal)

The contraction uses the **full** blocked tree action: add the old-face and electric terms to (23), including cross bubbles with their cubic vertices. Use the same gauge reduction, background and physical units on both sides of the cut. The following bounds concern one pristine step; all coefficients are bulk, zero-image perturbative coefficients.

First take a bounded background $\mathcal{B}$, with $b = \sup_e |\mathcal{B}_e|$, and set $z = L\mathcal{B}$, $h = -GT_y[z,z]/2$. At its conditional saddle the eliminated Hessian is

$$D(\epsilon) = H + \epsilon V_D + \epsilon^2 W_D + O(\epsilon^3), \qquad V_D = T_{yy}[z], \quad W_D = \tfrac{1}{2}U_{yy}[z,z] + T_{yyy}[h]. \tag{26}$$

The anchored bounds (25) and the Schur test give, with $d = 3(N^2-1)$,

$$\begin{gathered} v = Klb, \qquad w = \tfrac{1}{2}(K + gK^2)l^2 b^2, \\ \|D^{-1/2} V_D D^{-1/2}\|_{2,\tau} \le \sqrt{d}\, v/4, \qquad \|D^{-1/2} W_D D^{-1/2}\|_{1,\tau} \le dw/4, \\ |\tau(E_D - A_D^2/2)| \le d(w/4 + v^2/32). \end{gathered} \tag{27}$$

Here $D = H$ at zero background and τ is per mid-space site (one mid-space per slab). These are precisely Theorem 3's norms, uniform in volume and in an added nonnegative mass regulator. The saddle term $T_{yyy}[h]$ retains the exchange term in (23).

For clarity about the subtraction, let $\mathcal{H}$ be the pristine Hessian in retained/eliminated coordinates, evaluated on this saddle, and $S = \mathcal{H}/D$ the full blocked tree Hessian. Differentiation of the stationary action gives S exactly, including its cubic/quartic jets. With $\mathcal{C}(H) = \tau(E_H - A_H^2/2)$, determinant factorization at fixed regulator gives

$$\mathcal{C}(S) - \mathcal{C}(\mathcal{H}) = -\mathcal{C}(D). \tag{28}$$

In the mid-vertex gauge the eliminated gauge volume has already been removed. A background-gauge implementation must include its ghost and constraint determinants in (28); their net contribution is the same reduced integral. Local heat-kernel and Haar amplitudes are kept in m, as in (10). Subtract the **same** continuum determinant from both massless sides of (28), before taking their difference. Thus (27) bounds the determinant part of the subtracted one-step change. The retained massless A_S, E_S can separately have divergent norms; (28) controls their combined contraction and subtraction. Isolated diagrams built only from Γ_3, Γ_4 have no such assertion.

To extract F^2 with explicit constants, use the constant commuting flux tests of the one-step note, Theorem 1 (full-read: §§1–5). They are stationary interpolated backgrounds, so (28) applies without a background field redefinition. Its magnetic symbol includes the orbital terms needed for this extraction; (27) alone, with a bounded potential, would leave that issue unresolved. Here is an explicit bound on that note's integral $\mathcal{I}_{jk}$. In its (6)–(8), at $M = 4I, W = I/2$, put $a_* = 64$, $r = (4+v)^{-1}$. The absolute trigonometric coefficient row sums bound H_1, H_2,

the first two momentum derivatives of H_0, and the first momentum derivatives of H_1 by a_*. The inverse derivative identities give

$$\begin{aligned} \|\partial R_0\| &\le a_* r^2, & \|\partial^2 R_0\| &\le a_* r^2 + 2a_*^2 r^3, \\ \|R_1\| &\le a_* r^2 + a_*^2 r^3, & \|\partial R_1\| &\le a_* r^2 + 4a_*^2 r^3 + 3a_*^3 r^4, \\ \|R_2\| &\le a_* r^2 + \tfrac{7}{2} a_*^2 r^3 + 6a_*^3 r^4 + 3a_*^4 r^5. \end{aligned} \tag{29}$$

Each ∂ is either magnetic momentum derivative; mixed second derivatives obey the same bound. For example the two Poisson products cancel the factor $1/2$ in an absolute bound, while the four second derivative products give $4/8$ in the last term of (8). Integrating (29), using $|\operatorname{tr}_3 R_2| \le 3\|R_2\|$, gives

$$|\mathcal{I}_{jk}| \le D_* := 3\left(\frac{a_*}{4} + \frac{7a_*^2}{64} + \frac{a_*^3}{32} + \frac{3a_*^4}{1024}\right). \tag{30}$$

At $a_* = 64$ this explicit constant is $D_* = 173424$.

For a unit flux in any one plane the full normalized conditional one-loop coefficient $d_{\mu\nu} = 2A_{\mu\nu}$ therefore satisfies

$$|d_{1j}| \le N/48, \qquad |d_{jk}| \le N(1/24 + D_*) =: D_N. \tag{31}$$

These constants include the bridge normalization and the transverse heat-kernel amplitude $-N/24$ in the inverse coupling. In a common scheme at the same physical scale, (28), with these local terms, says $c^{\text{tree}}_{\text{blocked}} = c_{\text{pristine}} - d_{\mu\nu}$ for the selected plane coefficient. Thus its finite matching change is bounded by D_N ($3(1/24 + D_*)$ for $SU(3)$); in an arbitrary fixed scheme $|c^{\text{tree}}_{\text{blocked}}| \le |c_{\text{pristine}}| + D_N$. One may fix the common finite subtraction by $c_{\text{pristine}} = 0$. The pristine continuum-subtracted coefficient and Ward extraction are understood in the formal matching scheme of the one-step note. Keeping the generated one-loop local term adds $d_{\mu\nu}$ back, as required by exact integration. Changing the physical matching scale also adds the universal logarithm; (31) uses equal scales throughout.

Infrared term. The common massless F^2 logarithm has coefficient $2b_0 = 11N/(24\pi^2)$; it cancels in (28). The surviving transverse bridge density at $k = 0$ is $-N/24 - N \operatorname{Re} \int_0^\infty \operatorname{tr}_3 R_2(0, v)\, dv$. It controls the leading small-ball contribution, bounded by $D_N \varepsilon^3/(6\pi^2)$ for $|k| < \varepsilon \le 1$. The cut density vanishes quadratically and contributes at most $N\varepsilon^5/(2304\pi^2)$, since $Q \le |k|^2$ and $8 + Q \ge 8$. Equation (29) supplies domination through the infrared limit, including the orbital response. This proves finite one-step matching within the stated formal scheme; depth-uniform (14) remains the next estimate.

7.2. Uniform iteration requirements (conditional)

To iterate recursion (12) uniformly, each **updated** action, in rescaled lattice units, must have a conditional Hessian $D_n \ge \gamma I$ and anchored bounds (25) with common positive γ, α and finite g, K, l, including the constraint's background jets; anisotropies must remain bounded above and away from zero. Ghosts, Jacobians and generated one-loop local terms must be carried in the same Ward-compatible scheme. These hypotheses give bounds of the form (27)–(31) with common constants. Uniform endpoint matching requires (14) for the full retained, continuum-subtracted integrand, with regulator limits justified, for example, by a common integrable envelope after the F^2 derivatives. Bounded individual bridge contributions allow accumulation of order n; their scale-adjusted partial sums need separate control (sublinear control suffices for Theorem 4). Proving these bounds for the generated trajectory, and the interacting remainder bound (17), remains open; Theorem 2 excludes inferring them from a nondegenerate sharp Gaussian fixed point.

8. Consequence for STATE

The exact sharp-blocking obstruction and regulated Hessian-jet criterion are separate from the one-step coefficients in the matching treatment. The commonly subtracted formal one-step change is finite with explicit constants (24)–(31). Depth-uniform endpoint matching (14), the updated-action hypotheses of §7.2 and interacting remainder control remain unproved. None is supplied by Gaussian kernel convergence, which fails for the stated sharp map. Internal written checks are not independent review; new research remains suspended.